

6.7 Reparameterization, centring and R^2

6.7.1 Same subspace, same fit

The key idea of [Section 6.1](#) now becomes a theorem.

Theorem 6.7.1 — Invariance under reparameterization

Let $\mathbf{X}$ ($n \times p$) and $\mathbf{W}$ ($n \times q$) satisfy $\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{W})$. Then the two linear models $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $E(\mathbf{Y}) = \mathbf{W}\boldsymbol{\gamma}$ have the same projection $\mathbf{M}$, the same fitted values and residuals for every $\mathbf{y}$, and the same residual sum of squares. There are a $p \times q$ matrix $\mathbf{T}$ with $\mathbf{W} = \mathbf{X}\mathbf{T}$ and a $q \times p$ matrix $\mathbf{S}$ with $\mathbf{X} = \mathbf{W}\mathbf{S}$. If $\hat{\boldsymbol{\gamma}}$ is a least squares estimate for $\mathbf{W}$, then $\mathbf{T}\hat{\boldsymbol{\gamma}}$ is one for $\mathbf{X}$, and $\mathbf{S}\hat{\boldsymbol{\beta}}$ is one for $\mathbf{W}$ whenever $\hat{\boldsymbol{\beta}}$ is one for $\mathbf{X}$. If both matrices have full column rank, then $p = q$, $\mathbf{T} = \mathbf{S}^{-1}$, and $\hat{\boldsymbol{\gamma}} = \mathbf{S}\hat{\boldsymbol{\beta}}$.

Proof. A projection is determined by its range (Theorem 6.3.3), so both models have projection $\mathbf{M}$, and fitted values and residuals follow from Theorem 6.4.1. Each column of $\mathbf{W}$ lies in $\mathcal{C}(\mathbf{X})$, so $\mathbf{W} = \mathbf{X}\mathbf{T}$ for some $\mathbf{T}$, and likewise $\mathbf{X} = \mathbf{W}\mathbf{S}$. If $\mathbf{W}\hat{\boldsymbol{\gamma}} = \mathbf{M}\mathbf{y}$, then $\mathbf{X}(\mathbf{T}\hat{\boldsymbol{\gamma}}) = \mathbf{M}\mathbf{y}$, so $\mathbf{T}\hat{\boldsymbol{\gamma}}$ is a least squares estimate for $\mathbf{X}$. The other direction is symmetric. With full column rank, $\mathbf{X} = \mathbf{X}\mathbf{T}\mathbf{S}$ gives $\mathbf{X}(\mathbf{I} - \mathbf{T}\mathbf{S}) = \mathbf{0}$ and so $\mathbf{T}\mathbf{S} = \mathbf{I}$, and similarly $\mathbf{S}\mathbf{T} = \mathbf{I}$. Uniqueness of least squares estimates then gives $\hat{\boldsymbol{\gamma}} = \mathbf{S}\hat{\boldsymbol{\beta}}$. ■

Many operations that change the printed coefficient table leave the fit exactly unchanged:

- **Changing units** of a regressor (metres to centimetres, dollars to thousands of dollars) rescales one column: $\mathbf{W} = \mathbf{X}\mathbf{D}$ with $\mathbf{D}$ diagonal. The coefficient is divided by the scale factor, and nothing else changes.
- **Centring** regressors, $\mathbf{x}_j \mapsto \mathbf{x}_j - \bar{x}_j\mathbf{1}$, when the model has an intercept. The slopes are unchanged, and the intercept becomes the fitted value at the mean of the regressors instead of at zero, which is often far outside the data.
- **Recoding a factor**. Reference-level dummies, sum-to-zero effect coding, and cell-means coding without an intercept all span the space of vectors that are constant within levels.
- **Replacing polynomial terms by orthogonal polynomials** ([Chapter 42](#), Theorem 42.2.2), which span the same space with much better numerical behaviour ([Section 6.10](#)).

Operations that *do* change the fit are exactly those that change $\mathcal{C}(\mathbf{X})$. Examples are dropping the intercept when no other combination of columns produces $\mathbf{1}$, replacing a regressor by its logarithm, and adding an interaction.

6.7.2 Centring as a projection

Suppose $\mathbf{1} \in \mathcal{C}(\mathbf{X})$. By Theorem 6.5.1 with $\mathbf{X}_0 = \mathbf{1}$, the fitted vector splits as

$$\mathbf{M}\mathbf{y} = \bar{y}\mathbf{1} + (\mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{y},$$

and the second projection is onto $\mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp$, the vectors in the model space with entries summing to zero. By Lemma 6.6.1 with $\mathbf{X}_2 = \mathbf{1}$, that subspace is spanned by the centred columns $(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{X}$. Applying Pythagoras to the orthogonal pieces $\hat{\mathbf{Y}} - \bar{y}\mathbf{1}$ and $\hat{\mathbf{e}}$ of $\mathbf{y} - \bar{y}\mathbf{1}$ gives the *corrected* decomposition

$$\underbrace{\sum_i (y_i - \bar{y})^2}_{\text{SST}} = \underbrace{\sum_i (\hat{y}_i - \bar{y})^2}_{\text{SSR}} + \underbrace{\sum_i (y_i - \hat{y}_i)^2}_{\text{SSE}}, \quad (6.7.1)$$

with dimensions $n - 1 = (r - 1) + (n - r)$. Chapter 9 builds the analysis of variance on this identity.

6.7.3 R^2 is a squared cosine

The **coefficient of determination** of a model with $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ is

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}.$$

It depends on the model only through $\mathcal{C}(\mathbf{X})$, so by Theorem 6.7.1 it is invariant under every reparameterization. Its geometric meaning is simple.

Theorem 6.7.2

Assume $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{y} \notin \text{span}(\mathbf{1})$. Let θ be the angle between the centred response $\mathbf{y} - \bar{y}\mathbf{1}$ and the centred fitted vector $\hat{\mathbf{Y}} - \bar{y}\mathbf{1}$. Then

$$R^2 = \cos^2 \theta = \text{corr}(\mathbf{y}, \hat{\mathbf{Y}})^2,$$

where corr is the sample correlation. For simple regression on one regressor $\mathbf{x}$, $R^2 = \text{corr}(\mathbf{y}, \mathbf{x})^2$.

Proof. Write $\mathbf{u} = \mathbf{y} - \bar{y}\mathbf{1}$ and $\mathbf{v} = \hat{\mathbf{Y}} - \bar{y}\mathbf{1}$. Then $\mathbf{u} = \mathbf{v} + \hat{\mathbf{e}}$ with $\mathbf{v} \perp \hat{\mathbf{e}}$, so $\mathbf{u}^\top \mathbf{v} = \|\mathbf{v}\|^2$ and

$$\cos^2 \theta = \frac{(\mathbf{u}^\top \mathbf{v})^2}{\|\mathbf{u}\|^2 \|\mathbf{v}\|^2} = \frac{\|\mathbf{v}\|^2}{\|\mathbf{u}\|^2} = \frac{\text{SSR}}{\text{SST}}.$$

(If $\mathbf{v} = \mathbf{0}$ the angle is undefined, but $R^2 = 0$ and the correlation is taken to be zero.) Since $\hat{\mathbf{Y}}$ has mean $\bar{y}$ (Proposition 6.4.3(b)), $\cos \theta$ is the sample correlation of $\mathbf{y}$ and $\hat{\mathbf{Y}}$. In simple regression $\mathbf{v} = (S_{xy}/S_{xx})(\mathbf{x} - \bar{x}\mathbf{1})$ by Example (Intercept inside simple regression), a scalar multiple of the centred regressor. Rescaling a vector by a nonzero constant does not change a squared cosine, so $\cos^2 \theta$ is the squared correlation of $\mathbf{y}$ and $\mathbf{x}$. When $S_{xy} = 0$ both sides are zero. ■

So R^2 measures how closely the model space lines up with the centred data. Figure 6.7.1 draws the case $n = 3$. For the three-regressor model of Example (Poverty and murder rates), $R^2 = 0.6419$, meaning the centred response makes an angle of 36.8° with the model space. The simple regression on poverty alone has $R^2 = 0.3599$, the square of the correlation 0.5999.

inside $\mathbf{1}^\perp$: $R^2 = \cos^2 \theta = 0.617$

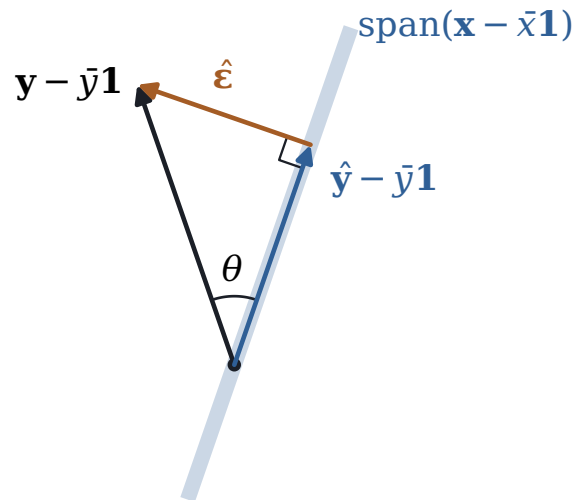


Figure 6.7.1: R^2 as a squared cosine, for $n = 3$ and one regressor. Centred vectors lie in the plane $\mathbf{1}^\perp$, so the whole configuration is two-dimensional. The centred fitted vector is the projection of the centred response onto the line spanned by the centred regressor.

```
import numpy as np
import statsmodels.api as sm

def r_squared_as_cosine(y, X):
    """X must contain the intercept column. Returns (R^2 from sums of squares, cos^2
    of angle)."""
    beta, *_ = np.linalg.lstsq(X, y, rcond=None)
    y_hat = X @ beta
    yc, yhc = y - y.mean(), y_hat - y.mean() # centre both vectors
    r2 = 1 - np.sum((y - y_hat) ** 2) / np.sum(yc ** 2)
    cos = (yc @ yhc) / (np.linalg.norm(yc) * np.linalg.norm(yhc))
    return r2, cos ** 2
```

[Run this code online](#)

Three consequences follow directly from the geometry.

1. R^2 never decreases when the model space grows. If $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ and both contain $\mathbf{1}$, then $\text{SSE}_0 - \text{SSE} = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 \geq 0$ by Eq. 6.5.2. Adding a regressor of pure noise increases R^2 almost surely. This is why R^2 cannot be used for model selection ([Chapter 29](#)).
2. $R^2 = 1$ once the model space contains $\mathbf{y}$, which always happens when $\text{rank}(\mathbf{X}) = n$. A model with as many free parameters as observations interpolates, and its R^2 says nothing.
3. Without an intercept, the definition changes. If $\mathbf{1} \notin \mathcal{C}(\mathbf{X})$, the corrected decomposition [Eq. 6.7.1](#) fails, because $\hat{\mathbf{Y}} - \bar{y}\mathbf{1}$ and $\hat{\mathbf{e}}$ need not be orthogonal. Software then usually reports the *uncentred* $R_0^2 = \|\hat{\mathbf{Y}}\|^2 / \|\mathbf{y}\|^2$, the squared cosine of the angle between $\mathbf{y}$ and $\mathcal{C}(\mathbf{X})$ itself. It is typically much larger and cannot be compared with the centred R^2 of a model with an intercept ([Exercise \(B1\)](#)).

6.7.4 Exercises

6.7.4.1 A. Check your understanding

Exercise A1.

A factor with three levels is coded with an intercept and indicators for levels 2 and 3. Write down the matrices $\mathbf{T}$ and $\mathbf{S}$ of [Theorem 6.7.1](#) that convert to the coding in which level 3, rather than level 1, is the omitted baseline, and to sum-to-zero effect coding. Express each new coefficient in terms of the old ones.

Exercise A2.

Show that, for a model with $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, R^2 is unchanged when $\mathbf{y}$ is replaced by $a\mathbf{1} + b\mathbf{y}$ with $b \neq 0$, and when any column of $\mathbf{X}$ is replaced by an affine function of itself. Is it unchanged when a regressor is replaced by its logarithm?

6.7.4.2 B. Practice

Exercise B1.

Show that if $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ then $R_0^2 = \|\hat{\mathbf{Y}}\|^2 / \|\mathbf{y}\|^2 = (n\bar{y}^2 + \text{SSR}) / (n\bar{y}^2 + \text{SST}) \geq R^2$. Construct a data set and a model without an intercept for which $R_0^2 > 0.95$ but the model's residual sum of squares exceeds that of the intercept-only model.

Solution. With $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, the fitted vector splits orthogonally as $\bar{y}\mathbf{1} + (\hat{\mathbf{Y}} - \bar{y}\mathbf{1})$, so $\|\hat{\mathbf{Y}}\|^2 = n\bar{y}^2 + \text{SSR}$, and likewise $\|\mathbf{y}\|^2 = n\bar{y}^2 + \text{SST}$. For $a \geq 0$ and $0 \leq \text{SSR} \leq \text{SST}$, $(a + \text{SSR}) / (a + \text{SST}) \geq \text{SSR} / \text{SST}$. For the example, take responses near 100 with small noise and a regressor unrelated to them, and fit through the origin. The fitted line captures the large mean, so R_0^2 is near 1, but its residuals exceed those around $\bar{y}$.

6.7.4.3 C. Going deeper

Exercise C1.

Suppose $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\mathbf{I})$ and $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ with $\text{rank}(\mathbf{X}) = r$. Using the results of [Chapter 4](#), show that $R^2 \sim \text{Beta}((r-1)/2, (n-r)/2)$, so that $E(R^2) = (r-1)/(n-1)$ even though no regressor is related to the response. Use this to motivate the adjusted $R^2 = 1 - (1 - R^2)(n-1)/(n-r)$.